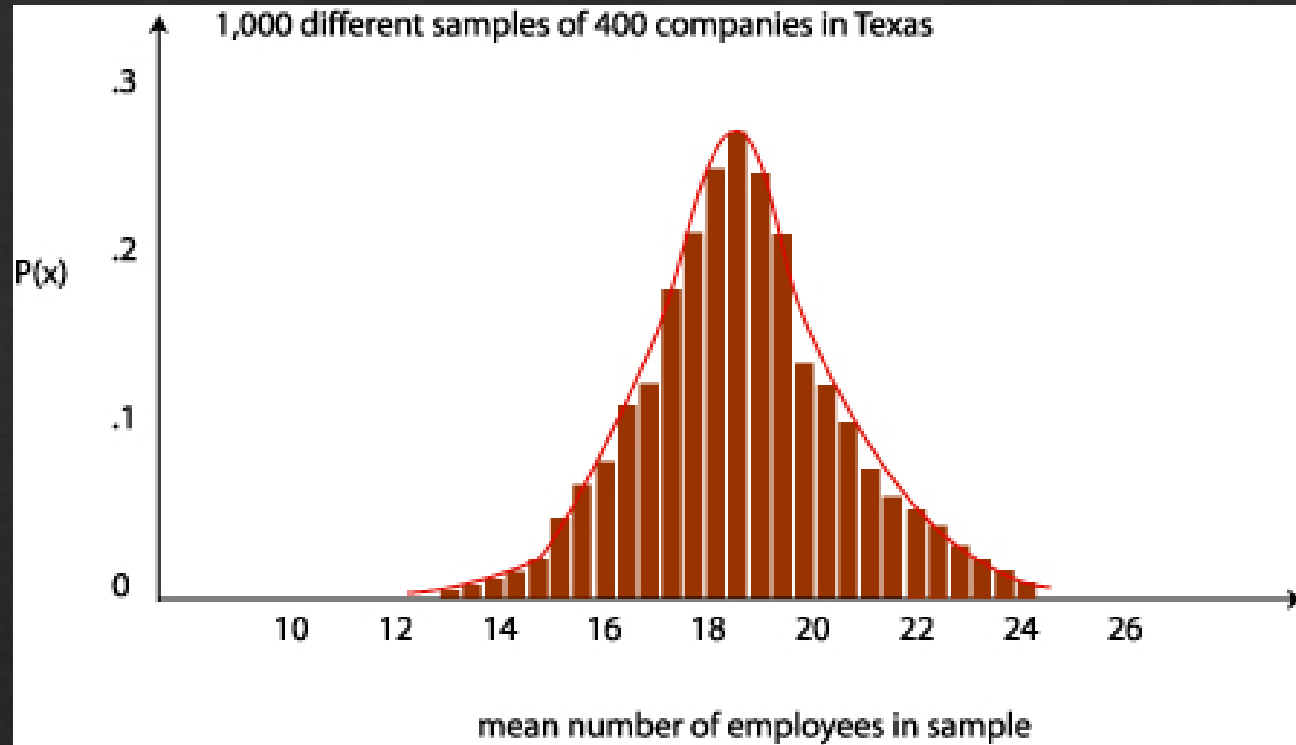




# Week 6: Central limit theorem and confidence intervals

Geog4300: Shannon

# This week

- Z/t scores
  - Central limit theorem
  - Confidence intervals
  - Sampling
- 
- Lab 3 will be available after class on Thursday. Due date: October 23 (Thursday)
  - Thoughts/reflections from Lab 2?

# Z (and t) scores

How many standard deviations is an observation from the mean?

$$Z_x = \frac{X - \mu_X}{\sigma}$$

These can also be called standardized anomalies.

$$Z_x > 1$$

$$Z_x < -1$$

How would you interpret these values?

# Z (and t) scores

How many standard deviations is an observation from the mean?

$$Z_x = \frac{X - \mu_X}{\sigma}$$

For a variable with a mean of 23 and a s.d. of 4, the z-score for 30 would be:

$$\frac{30 - 23}{4} = 1.75$$

What value would produce a z-score of negative 2?

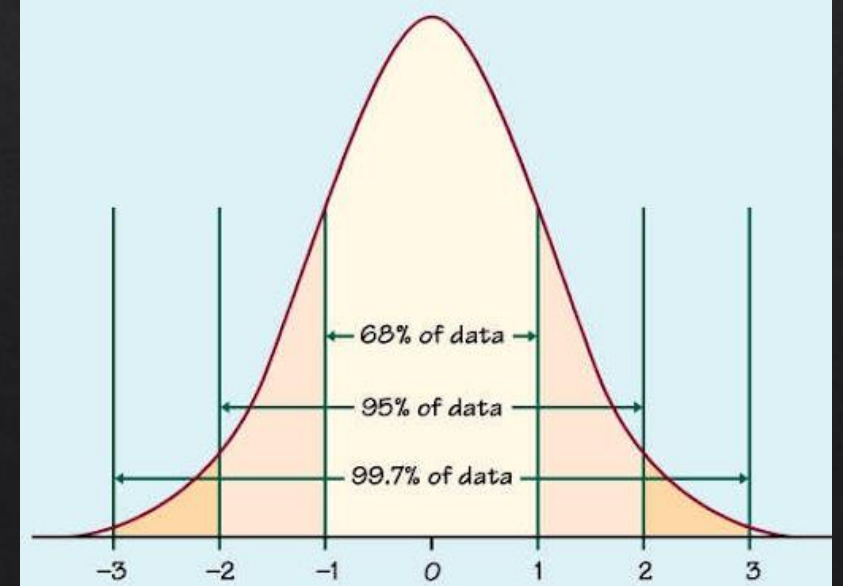
# Z (and t) scores: Why do they matter?

- 1) They provide an easy way to compare data from different distributions (e.g. daytime and nighttime temperatures)
- 2) In a normal distribution, we can identify what percentage of observations fall within given range of Z scores.

Z scores of **-1 to 1**: 68% of observations

Z score of **-1.75 to 1.75**: 90% of observations

Z score of **-1.96 to 1.96**: 95% of observations



# Data challenge!

Pick a variable from our county census dataset. Transform its values to z-scores in Excel. You'll need to:

- 1) Calculate the mean and standard deviation for all observations
- 2) Subtract the mean and divide by standard deviation for all observations

How many counties are above 2? How many below -2? In what states/regions are these located?

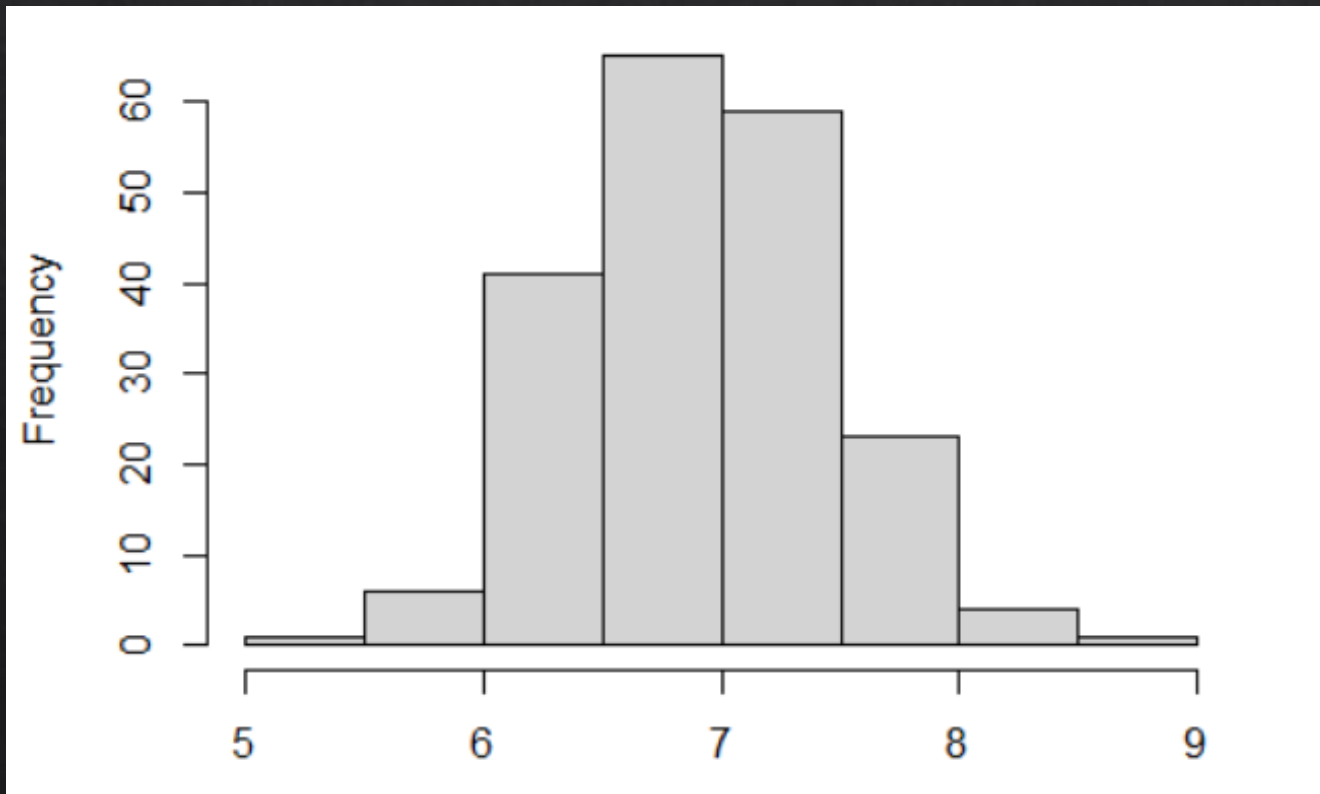


# The **central limit theorem**:

What's the relationship between **the mean** of the sample population and the actual population?



Central limit theorem: The distribution of sample means is *normally distributed* and centers around the *population mean*.



The actual mean is here *somewhere!*





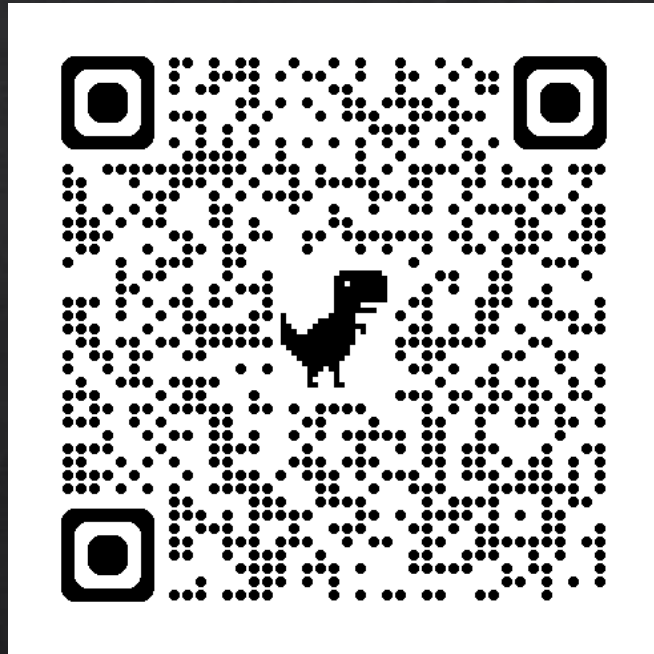
# The Central Limit Theorem...



...Clearly Explained!!!

# Let's roll!

Roll your die 10 times. Put the results on this page:



Then do two more sets of 10 rolls (30 total rolls)

# Identifying our terms

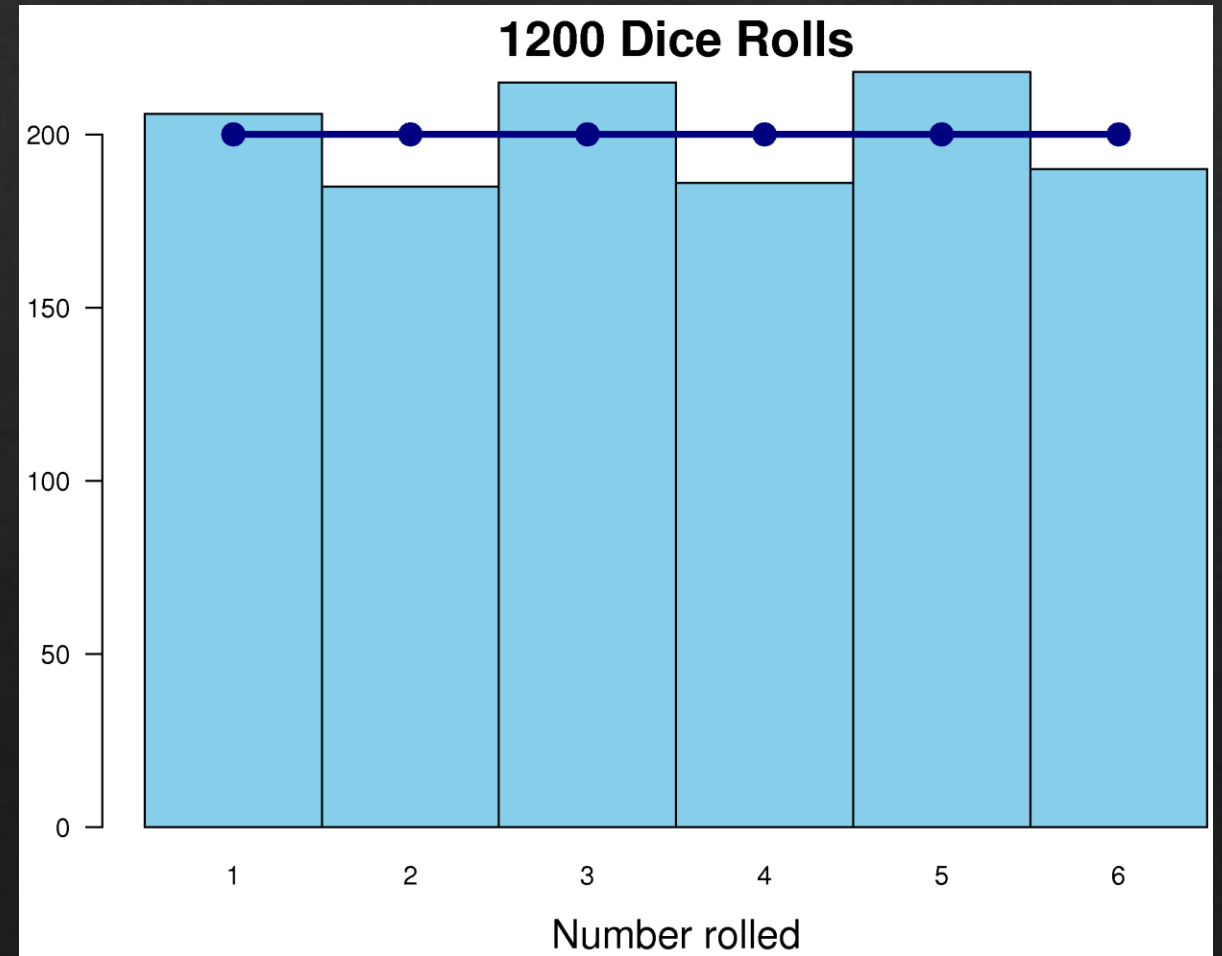
- **Samples:** Each individual set of 10 dice rolls
- **Sample mean distribution:** Just the mean values for each set of 10 dice rolls
- **Population:** All possible means for of 10 dice rolls in the universe

# What's the mean of the population?

Mean of two dice rolls: 3.5

Standard deviation: 1.7

How did the means of your three  
“samples” compare?

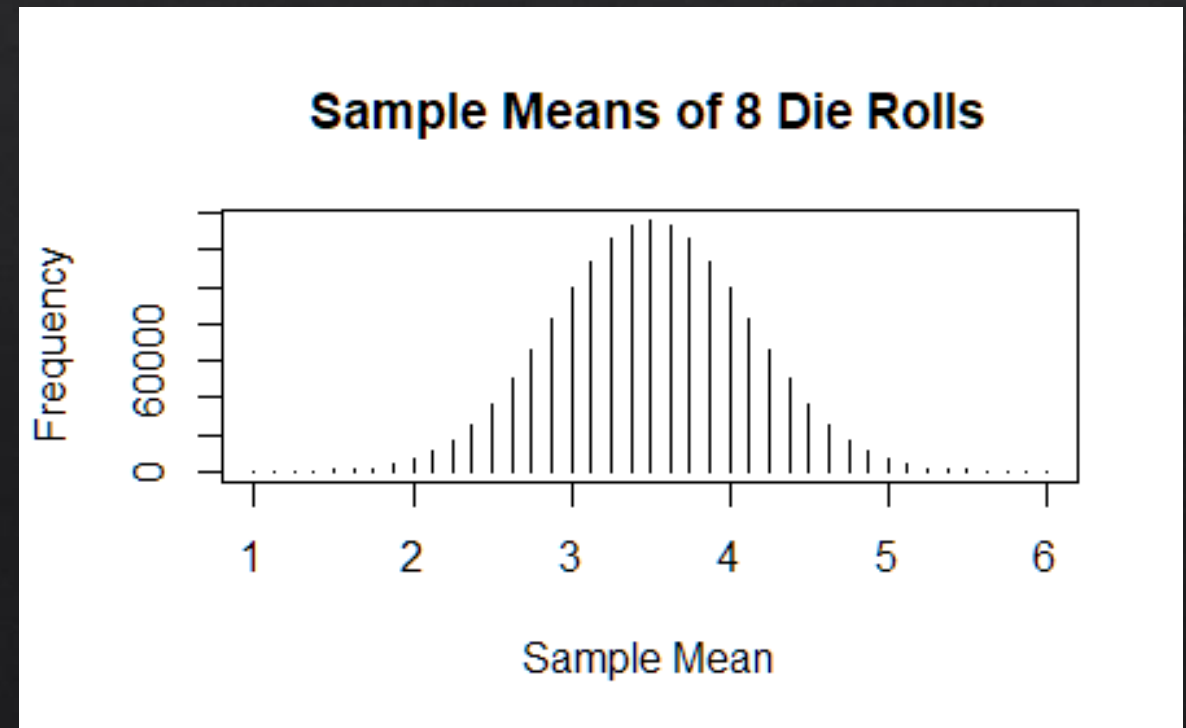




What's the **mean** and **standard deviation** for our whole class's rolls (the sample mean distribution)?

Mean of these means: 3.5

Standard deviation of these means: 0.5



# Standard error

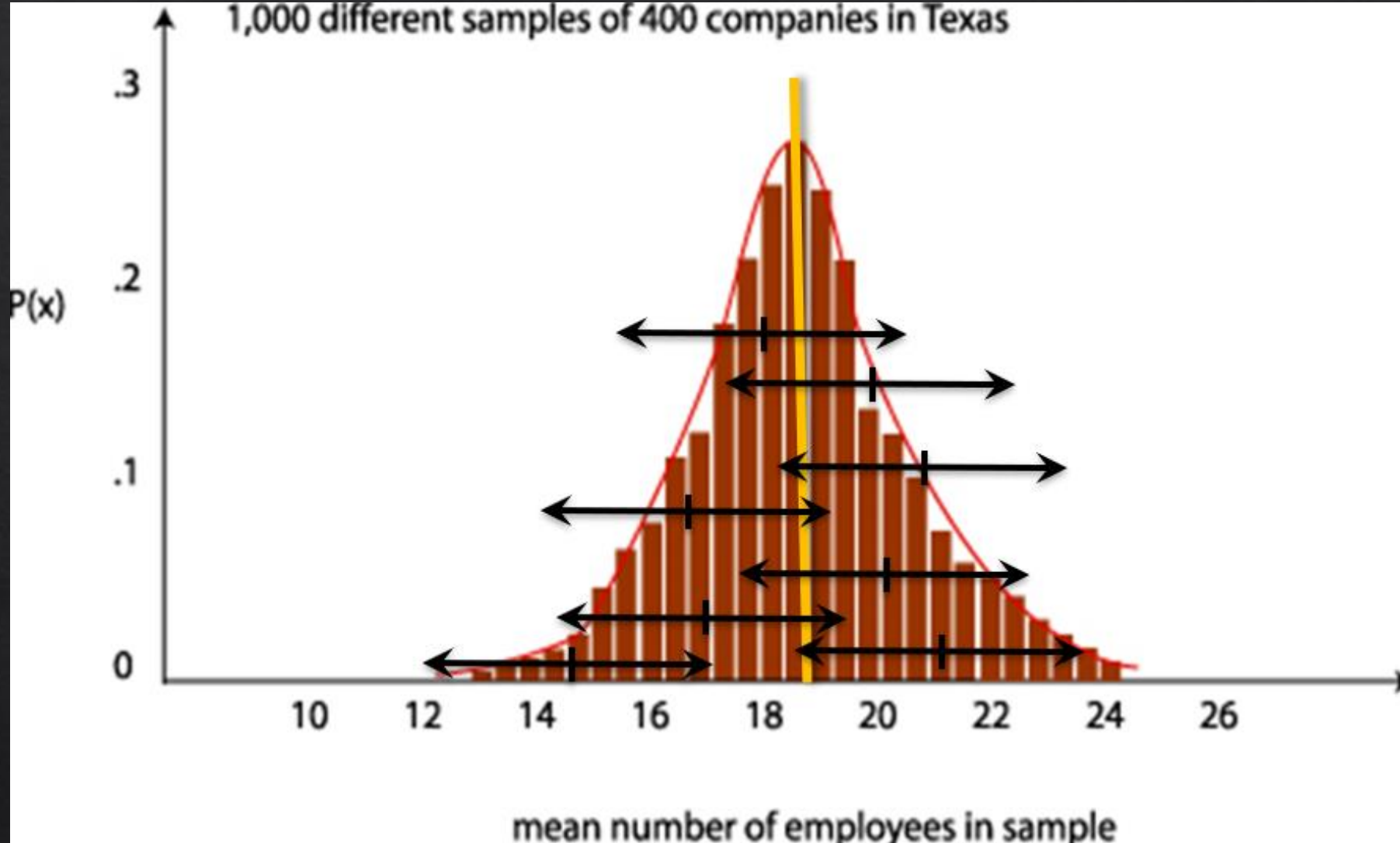
$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$\sigma$ =standard deviation  
of sample or population

How many 10 dice roll samples  
did we create? What should our  
standard error be?

How many means fit within 1  
standard error of the mean?  
Within 2 standard errors?

# Confidence intervals



# Confidence intervals

Three ingredients for a confidence interval:

- Sample mean
- Standard error
- Z-score for the confidence interval

Margin of error = z-score \* the standard error  
Subtract and add this MOE to the sample mean

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

What's the confidence interval for each of your three dice roll samples? How many include the actual mean of 7?



# Confidence intervals for proportions

You asked 50 households in Athens if they keep chickens on their property. Three said they did. Create a confidence interval for the proportion of chicken owning households in the city.



$$SE_p = \sqrt{\frac{p(1-p)}{n-1}}$$



# Let's talk about sampling!





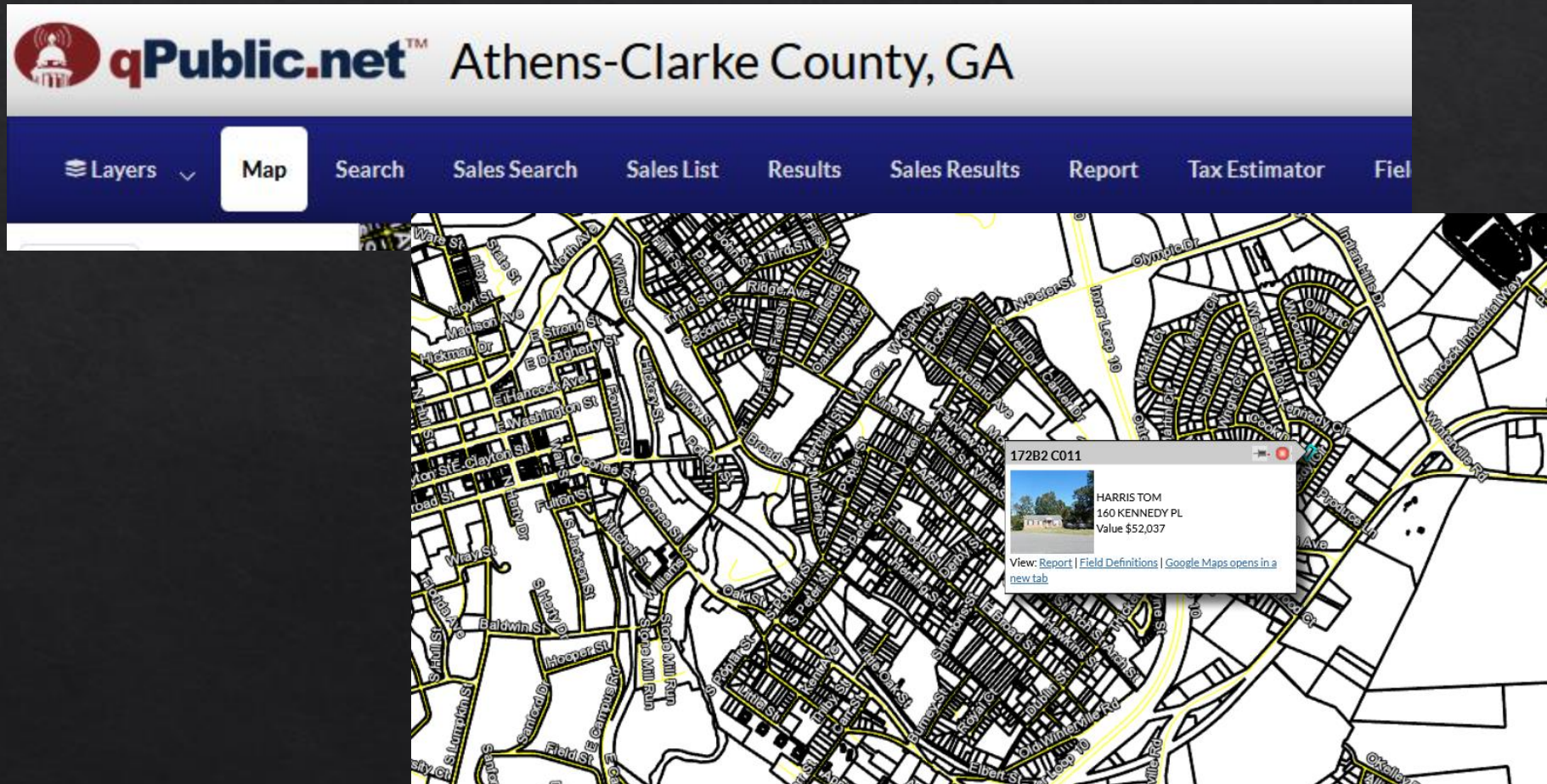
# Sample research question:

What are the most significant obstacles homeowners face in maintaining their homes? How does this vary by the homeowners' age?



# Sampling frame:

List of ALL possible households you could interview





# Sampling strategy

## Non-probability sample

Convenience

Snowball

Quota

## Probability sample

Random

Systematic

Stratified

Cluster



# Sample size

If the MOE for proportions is

$$MOE = Z * \sqrt{\frac{p(1-p)}{n}}$$

Then we can compute sample size as

$$n = \frac{Z^2}{MOE^2} * (p * (1-p))$$

We asked 20 homeowners if they were worried about losing their homes in the next year. Four said yes. What's the confidence interval?

How big a sample would we need to get a margin of error of less than 1.5%?